ION CLUB Karpathos — No-Code Prototype Brief

A Glide-based NFC rental system for staff to pilot in one season before committing to a full custom build.

---

1. Context

ION CLUB Karpathos wants to introduce an NFC-based system that tracks who is on the water, what equipment they're using, and the wind conditions during their session. The long-term goal is a custom-built application. Before committing €25,000–50,000 to that, the station wants to validate the concept with a no-code prototype built in Glide (with AppSheet as a fallback option).

The prototype needs to cover:

1. **Safety** — live dashboard of who is on the water and for how long.
2. **Equipment tracking** — usage hours and session counts per board and sail.
3. **Post-holiday guest emails** — summaries with sessions, equipment used, and wind.

Scope for tagging: boards and sails only. Masts and booms are excluded to simplify the workflow.

---

2. Why Glide

Glide is the best fit for the use case because it has native NFC support on Android, handles offline operation gracefully (important on unreliable beach WiFi), produces genuinely good-looking interfaces, and connects easily to Google Sheets as a backend.

AppSheet is the fallback if concurrent multi-user editing becomes a problem — it handles simultaneous edits more robustly, but its UI is noticeably less polished.

---

3. Backend structure

The backend is Google Sheets, with five worksheets that mirror the intended production database.

Guests

Column	Notes
GuestID	Auto-generated
FirstName, LastName	
Email	For post-holiday summary

Column	Notes
Phone	
EmergencyContact	Name + phone
SkillLevel	beginner / intermediate / advanced / expert
WaiverSigned	Boolean
WaiverDate	
Nationality	ISO country code
LanguagePreference	For email
MarketingConsent	GDPR requirement
WristbandUID	Currently assigned NFC band
Notes	Free text

Equipment

Column	Notes
EquipmentID	Auto-generated
Type	board / sail
Brand, Model	
Size	e.g. 135 (litres) or 4.7 (m^2)
SizeUnit	litres / m^2
SkillRequired	beginner_ok / intermediate / advanced
Status	available / in_use / maintenance / retired
TagUID	NFC tag UID
TotalSessions	Counter
TotalHours	Counter
LastMaintenance	Date
Notes	

Sessions

Column	Notes
SessionID	Auto-generated
GuestID	Foreign key
BoardID	Foreign key
SailID	Foreign key
CheckOutTime	Timestamp
CheckInTime	Empty until return
Duration	Computed
WindAtCheckout	Reference to WeatherLog
StaffMember	Who processed check-out
Status	active / completed / overdue / incident
Notes	

WeatherLog

Column	Notes
Timestamp	
WindSpeed	Knots
WindGust	Knots
WindDirection	Degrees
AirTemp	°C
WaterTemp	°C
ConditionsSummary	e.g. "Meltemi N 22–28kt, gusting 32"
Source	api / manual / station

Staff

Column	Notes
StaffID	
Name	
Email	Used for Glide login
Role	instructor / manager / rental_desk
Active	Boolean

Relationships between tables are handled by Glide's **Relations** feature, not SQL joins.

---

4. App screens

Screen 1 — Dashboard (home)

Purpose: at-a-glance situational awareness for any staff member walking up to the tablet.

Elements:

- Live count of guests currently on the water (Sessions where `CheckInTime` is empty).
- Prominent **"New check-out"** button.
- Secondary **"Quick scan"** button that launches NFC scan immediately and routes based on what is scanned (wristband → guest detail; board/sail → equipment detail).
- Sunset countdown widget.
- Red alert banner if any active session has been open more than 3 hours.
- Current wind conditions (latest WeatherLog row).

Screen 2 — New check-out (3-step wizard)

Purpose: create a Session record with two linked pieces of equipment and weather context. Target time: 8–10 seconds.

- **Step 1 — Wristband scan.** NFC tap looks up guest by `WristbandUID`. Displays guest photo, name, skill level, emergency contact. Blocks continuation if guest already has an active session.
- **Step 2 — Board scan.** NFC tap looks up board by `TagUID`. Displays brand, model, size, maintenance status. Blocks continuation if board is in `maintenance` or `retired`. Warns (but does not block) if board's `SkillRequired` exceeds guest's `SkillLevel`.

- **Step 3 — Sail scan.** Same pattern as board.
- **Confirm button.** Creates a Session row with `CheckOutTime = now()`, links to the latest WeatherLog row, sets `Status = active`, and marks both equipment items as `in_use`.

Screen 3 — Check-in

Purpose: close an active session.

- Single NFC scan of wristband.
- App finds the active Session for that guest, sets `CheckInTime = now()`, computes `Duration`.
- Updates equipment counters: `TotalSessions += 1`, `TotalHours += duration`.
- Flips equipment `Status` back to `available`.
- Optional prompt: "Any equipment issues to report?" with a text field that appends to the relevant equipment's `Notes`.

Screen 4 — Active sessions (safety dashboard)

Purpose: live view of everyone on the water, optimised for glanceability.

- List of all active Sessions, sorted by `CheckOutTime` ascending (longest out first).
- Row colour coded by elapsed time: green (<2h), amber (2–3h), red (>3h).
- Each row shows: guest name, avatar, equipment, duration, skill level.
- Tap a row to reveal emergency contact and a **"Call guest"** button that launches the phone dialer.
- Filter: show only alerts (>3h out).

Screen 5 — Equipment browser

Purpose: inventory overview and manual status management.

- List of all boards and sails with filters: available / in_use / maintenance / all.
- Sort by total hours (for retirement decisions) or by last maintenance date.
- Tap an item to see its full session history.
- **"Mark for maintenance"** action flips status and removes from check-out flow.

Screen 6 (optional) — Guest browser

Purpose: look up a guest by name to see their history, or register a new guest.

- Search by name or email.
- Shows current stay, lifetime sessions, last visit date.
- Action: **"Assign new wristband"** — scan a band and write it to the guest's record.

- Action: **"Create new guest"** — form with all required fields.

---

5. What Glide handles natively

- NFC tag reading (UID lookup via Android NFC intent).
- User authentication via email whitelist.
- Action logging (automatic timestamps and staff attribution on all writes).
- Offline operation with automatic sync.
- Basic computed columns ("sessions this week", "total hours", etc.).
- Push notifications.
- Role-based access (instructors vs. managers see different screens).

---

6. What Glide does NOT handle — workarounds needed

Writing NFC tags

Glide reads tags but cannot write to them. Use the free **NFC Tools** app on Android to pre-encode each tag with its UID before attaching it to equipment. This is a one-time setup action per tag, about 30 seconds each.

Automated weather ingestion

No direct Glide integration. Two options:

- **Manual:** staff tap a "Wind update" button in the app every 1–2 hours, which logs a WeatherLog row with current conditions.
- **Automated:** set up a **Make.com** (formerly Integromat) scenario that calls Open-Meteo every 30 minutes and appends a row to the WeatherLog sheet. ~€20/month. This is the recommended approach.

Rich guest summary emails

Glide's built-in email actions are basic. For the designed visual email (hero image, stats, wind chart, equipment list):

- Build an HTML template in **Mailchimp** or **SendGrid**.
- Use **Make.com** to: (a) find stays where `departure_date < today - 3` and `summary_sent IS NULL`, (b) aggregate the guest's session data, (c) populate the email template, (d) send, (e) stamp `summary_sent_at` back to the sheet.
- One-time setup: ~1 day. Monthly cost: €20 Make.com + Mailchimp free tier for <500 emails/month.

Complex analytics

For season-over-season or heatmap-style analysis, export to Google Sheets or Google Data Studio. Glide's charts are basic but adequate for daily operational views.

Scale ceiling

Glide begins to lag around 25,000 rows per table. Expected load: 500 guests × ~20 sessions = 10,000 sessions/season. Good for 2 full seasons before archiving is needed.

---

7. Hardware for the pilot

- **2 Android tablets** with NFC (mid-range, in waterproof cases) — one at the rental desk, one mobile on the beach. ~€400–600 total.
- **Small NFC tag order** — 50 adjustable silicone wristbands + 100 equipment tags (disc + sewable PPS combo). ~€200–300.
- **1 station anemometer** (optional, if not using Open-Meteo): ~€150.

Total hardware for pilot season: **€750–1,050**.

---

8. Software cost per month

Service	Purpose	Cost
Glide Team plan	App platform	~€50/month
Make.com	Weather fetch + email automation	~€20/month
Mailchimp	Summary emails	Free (<500/mo)
Google Workspace	Backend (Sheets)	Existing / free
Total		**~€70/month**

---

9. Build timeline

Scenario	Time to working app
Experienced Glide developer, focused	3–5 days

Scenario	Time to working app
Developer new to Glide, focused	7–10 days
Glide freelancer on Upwork (budget €500–1,500)	~2 weeks

---

10. Recommended pilot rollout

Month 1 — Safety loop only. Just wristbands, just check-out/check-in, just the active sessions dashboard. No equipment tagging yet. No summary emails. Goal: staff comfort with NFC workflow, confirm offline behaviour, measure scan time.

Month 2 — Add equipment. Tag boards and sails. Add equipment screen. Add usage counters. Goal: validate that staff reliably scan all three items (wristband + board + sail) under time pressure.

Month 3 — Add summary emails. Enable the automated Make.com pipeline. Send summaries to departed guests. Measure open rates and replies. Goal: validate guest reception.

End of season — decide. Based on: staff adoption rate, guest feedback on wristbands, and data quality, decide whether to (a) continue on Glide for another season, (b) commission the custom build, or (c) adopt an off-the-shelf rental platform like Booqable with NFC extensions.

---

11. Known risks and mitigations

Risk	Mitigation
Staff skip scan steps when busy	Design flow so `Confirm` button is disabled until all three scans complete. Measure completion rate in first week and iterate.
NFC tag falls off sail	Sew PPS tokens into clew reinforcement patch; keep 10% spare tags on hand.
WiFi drops during busy check-out rush	Glide's offline queue handles this; verify sync behaviour under simulated network loss before going live.
Guest loses wristband	"Reassign wristband" action takes 10 seconds; charge a small replacement fee (~€5) to discourage casual loss.
Weather API outage	Fallback to manual wind entry button. Any session during the outage gets linked to the most recent WeatherLog row.
GDPR: guest data retention	Set up quarterly automated purge of inactive guest records; documented retention policy; consent flow at check-in for marketing emails.

Risk	Mitigation
Scale: Glide slowdown after 2 seasons	Archive sessions older than 18 months to a separate sheet; plan for platform migration by year 3.

---

12. Out of scope for the prototype

- Integration with existing booking systems (handled manually for the pilot).
- Payment processing.
- Lessons booking and instructor scheduling.
- Guest-facing app (guests interact only via wristband and the post-holiday email).
- Mast and boom tracking.
- Multi-language UI (English only for the pilot; guest emails can be templated per language later).

---

13. Success criteria for the pilot season

- ⩾90% of sessions have complete data (guest + board + sail + check-out + check-in).
- Zero safety incidents where staff were unaware a guest was still on the water.
- Scan-to-confirm time consistently under 15 seconds.
- Staff self-report the system is faster than the previous paper/memory process.
- ⩾40% open rate on the summary emails.
- ⩾5 direct booking inquiries for next season traceable to the summary emails.

---

14. Supporting documents

- Database schema (PostgreSQL): `ion_club_rental_schema.sql`
- UI mockup (HTML): `ion_club_rental_app_mockup.html`

Prepared for ION CLUB Karpathos, 2026 season pilot planning.